A Web-Based Application for Local Deals and Discounts

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

Author:	**Zunaira Kanwal**
Roll No:	**F23BINFT1M01082**
Department:	**DEPARTMENT OF INFORMATION TECHNOLOGY**
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Document Approval

Daily Deals & Discount — The following Software Requirements Specification has been accepted and approved by:

Signature	Printed Name	Title	Date
	Dr. Muhammad Dost Khan	Supervisor / Chairman	30.10.2026

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1. Introduction

Consumers often struggle to find reliable local deals from nearby businesses. Local vendors also lack simple platforms to promote special offers digitally. The Daily Deals and Discounts web platform solves this problem by connecting customers and vendors in one virtual marketplace where real-time deals can be posted, viewed, and filtered easily.

1.1 Purpose

The purpose of this project is to build a web system that allows:

- Vendors to post deals, discounts, and promotional offers.
- Customers to browse, search, and locate nearby deals.
- The system to ensure validity, visibility, and user engagement.

1.2 Scope

Included Features:

- User registration & login (Vendor & Customer).
- Vendor dashboard to add, edit, and manage deals.
- Customer interface to browse deals by category, location, or search keywords.
- Geo-location-based deal suggestions.
- Ratings & feedback on deals.
- Admin panel for moderation and user management.

Excluded Features: Online payment system, delivery or order booking functionality, vendor financial statistics beyond deal analytics.

1.3 Definitions, Acronyms, and Abbreviations

Definitions:

- **Deal:** A promotional offer, discount, or special price provided by a vendor.
- **Vendor:** A registered business user who creates and manages deals on the platform.
- **Customer:** A user who searches, views, saves, and reviews available deals.
- **Admin:** System administrator responsible for approving deals and managing users.
- **Dashboard:** User interface where vendors or admins perform management activities.
- **Geo-location:** Technology used to identify and suggest deals based on location.

Acronyms & Abbreviations:

SRS: Software Requirements Specification | **FR:** Functional Requirement | **NFR:** Non-Functional Requirement | **UC:** Use Case | **UI:** User Interface | **DB:** Database | **HTML:** HyperText Markup Language | **CSS:** Cascading Style Sheets | **JS:** JavaScript | **VPS:** Virtual Private Server | **Pwd:** Password | **Img:** Image | **Mgmt:** Management

1.4 References

- PakWheels Classifieds platform architecture model
- OLX Pakistan local marketplace model
- Google Maps API Documentation

2. The Overall Description

The Daily Deal and Discounts system is a web-based platform designed to help users discover the best local deals, discounts, and offers available in their nearby area. Local vendors can post deals for products and services, while customers can search, filter, and locate these offers using categories, location, and price ranges. This SRS outlines system purpose, scope, functional, and non-functional requirements.

2.1 Product Perspective

This system is a standalone web application consisting of three modules: Admin, Vendor, and Customer.

2.1.1 Operations

- User registration, authentication, and password management.
- Vendor deal posting, updating, and image management.
- Customer deal searching, filtering, details viewing, saving, and reporting.
- Admin approval/rejection of vendor accounts and posted deals.
- Location-based deal suggestions, ratings, and feedback handling.

2.1.2 Site Adaptation Requirements

- Mobile-responsive design across modern web browsers (Chrome, Edge, Firefox, Safari).
- Automatic layout adjustment according to device resolution and active internet connectivity.

2.2 Product Functions

Standalone web application providing centralized management tools for Admin, Vendor, and Customer personas.

2.3 User Characteristics

User Type	Technical Level	Responsibilities
Admin	Intermediate	Approving deals, managing users, monitoring platform activities.
Vendor	Basic	Posting offers, updating deals, viewing offer analytics.
Customer	Basic	Searching, filtering, saving, and reviewing nearby deals.

2.4 General Constraints

- Requires stable internet connectivity and modern web browser support.
- Admin-only permissions for deal/vendor approvals; secure password hashing required.

2.5 Assumptions and Dependencies

- Users have internet-enabled devices and provide valid registration details.
- Hosting server uptime and user-granted location permissions for geo-location features.

3. Specific Requirements

3.1 Functional Requirements

Req ID	Description
FR-01	The system shall allow users to create accounts using personal details.
FR-02	The system shall allow vendors to log in using registered email and password.
FR-03	The system shall allow customers to log in securely to access deals.
FR-04	The system shall authenticate all users through secure login validation.
FR-05	The system shall allow vendors to post new deals with product details.
FR-06	The system shall allow vendors to upload product images while creating deals.
FR-07	The system shall allow vendors to set discount percentages and expiry dates for deals.
FR-08	The system shall allow customers to search deals using keywords.
FR-09	The system shall allow customers to filter deals by category, location, and price range.
FR-10	The system shall display filtered deals based on customer search criteria.
FR-11	The system shall allow admins to approve or reject vendor accounts.
FR-12	The system shall allow admins to approve or reject posted deals.
FR-13	The system shall allow admins to activate or deactivate user accounts.
FR-14	The system shall allow vendors to manage and update their posted deals.
FR-15	The system shall allow customers to view detailed information about deals.

3.2 Use Cases Summary

UC ID	Use Case Name	Actor(s)	Key Objective
UC-001	Sign Up	Customer, Vendor	User account creation with input validation.
UC-002	Login	All Users	Secure authentication and role-based redirect.
UC-003	Post a Deal	Vendor	Create deal, upload image, and submit for review.
UC-004	Search & Browse	Customer	Filter deals by location, category, and price.
UC-005	View Deal Details	Customer	Display complete deal info, vendor, and expiry date.
UC-006	Save or Report Deal	Customer	Bookmark deal or report inappropriate content.
UC-007	Approve/Reject Deal	Admin	Review and update deal publication status.
UC-008	Approve/Reject User	Admin	Manage and verify vendor/user account status.

3.3 Classes / Objects

1. User Object
• Attributes: UserID, FullName, Email, PasswordHash, PhoneNumber, Location, UserRole
• Functions: SignUp(), Login(), UpdateProfile()

2. Vendor Object
• Attributes: VendorID, UserID, BusinessName, ApprovalStatus
• Functions: PostDeal(), EditDeal(), DeleteDeal(), ViewAnalytics()

3. Customer Object
• Attributes: CustomerID, UserID, SavedDeals, SearchHistory
• Functions: SearchDeals(), ViewDeal(), SaveDeal(), ReportDeal(), ReviewDeal()

4. Deal Object
• Attributes: DealID, Title, Description, DiscountPercentage, OriginalPrice, DiscountedPrice, Location, ExpiryDate, Image, Status
• Functions: CreateDeal(), UpdateDeal(), DisplayDeal(), DeleteDeal()

5. Admin Object
• Attributes: AdminID, Username, PasswordHash
• Functions: ApproveDeal(), RejectDeal(), ManageUsers(), MonitorActivities()

6. Review Object
• Attributes: ReviewID, CustomerID, DealID, Rating, Comment, Date
• Functions: AddReview(), EditReview(), DeleteReview()

7. Report Object
• Attributes: ReportID, CustomerID, DealID, Reason, ReportDate, Status
• Functions: SubmitReport(), UpdateReportStatus()

8. Notification Object
• Attributes: NotificationID, UserID, Message, Date, Status
• Functions: SendNotification(), MarkAsRead()

9. Search Filter Object
• Attributes: FilterID, Keyword, Category, Location, PriceRange
• Functions: ApplyFilter(), ResetFilter()

10. Saved Deal Object
• Attributes: SavedDealID, CustomerID, DealID, SavedDate
• Functions: SaveDeal(), RemoveSavedDeal(), ViewSavedDeals()

3.4 Non-Functional Requirements

- **NFR001 - Performance:** The system should load deals within 3 seconds and support up to 500+ concurrent users seamlessly.
- **NFR002 - Security:** Password hashing must be implemented along with input validation for all input forms to prevent common web vulnerabilities.
- **NFR003 - Usability:** A highly intuitive, user-friendly, and mobile-responsive interface across all major browsers.
- **NFR004 - Reliability:** The system operational uptime should be at least 99% during active business hours.
- **NFR005 - Scalability:** The database must be optimized to handle expanding volumes of vendor and customer records over time.